

迷宮最短逃脫路徑 (佔分 20 分)

問題描述：

給定一個迷宮地圖，S 代表起點，E 代表終點，# 代表牆壁，. 代表可以走的路。請找出從 S 到 E 的最短路徑長度（所需的最少步數）。如果無法到達終點則輸出 -1。

玩家每次只能上下左右移動一格，不能穿過牆壁，也不能走出迷宮範圍。

輸入說明：

第一行：兩個整數 n m（迷宮的列數與行數， $n \leq 100$, $m \leq 100$ ）

接下來 n 行：迷宮地圖（每個位置只包含 S、E、#、. 其中一種）

輸出說明：

輸出一個整數：從 S 到 E 的最短步數，若無法到達則輸出 -1。

範例 1：

輸入：

```
5 5
S . . . .
. # . # .
. . . . .
# . # . E
. . . . .
```

輸出：

7

範例 2：

輸入：

```
5 5
S # # # .
# # . # #
# . . . #
# # # # E
. . . . .
```

輸出：

-1

Shortest Escape Path in a Maze (20 points)

Problem Description

You are given a maze represented as a grid. S denotes the starting position, E denotes the exit, # represents walls, . represents open paths. Your task is to find the shortest path length (minimum number of steps) required to move from S to E. If the exit cannot be reached, output -1.

The player can move one step at a time in four directions only: Up, Down, Left, Right. The player cannot pass through walls and cannot move outside the maze boundaries.

Input Description

The first line contains two integers n and m ($n \leq 100$, $m \leq 100$), representing the number of rows and columns of the maze.

The next n lines describe the maze grid. Each cell contains exactly one of the following characters: S, E, #, .

Output Description

Output a single integer: The minimum number of steps required to move from S to E. If it is impossible to reach the exit, output -1

Example 1 :

Input:

```
5 5
S . . . .
. # . # .
. . . . .
# . # . E
. . . . .
```

output :

7

Example 2 :

Input:

```
5 5
S # # # .
# # . # #
# . . . #
# # # # E
. . . . .
```

output :

-1